M2795.007800
Guidance, Navigation, and Control System
for Reusable Launch Vehicles

Project 2.
Development of Vehicle Simulator with Navigation Blocks



Project 2 Overview

- Project 2
  - Vehicle Simulator & Navigation for Reusable Launch Vehicles

- Objective
  - Develop a simulation model of Falcon 9
  - Implement the vehicle equations of motion and major environmental models
  - Reproduce representative flight phases using numerical simulation
  - Develop a navigation algorithm to estimate vehicle states from sensor measurements
  - Prepare a simulation environment for Guidance and Control design in Projects 3–4

- **Key Question**
  - Can we build a simulator that realistically represents how the vehicle moves and how its onboard system knows where it is?

Why This Project?

- From System Understanding to Mathematical Modeling
- Before designing Guidance & Control algorithms, we need a simulato r answers:
  - How does the vehicle move under forces and moments?
  - How do thrust, aerodynamics, and gravity affect the trajectory?
  - How does vehicle mass change during flight?
  - What measurements are available to the onboard computer?
  - How accurately can position, velocity, and attitude be estimated?
- Course Project Flow
  - P1. Vehicle & GNC Reverse Engineering → P2. Vehicle Simulator & Navigation → P3. Guidance Design → P4. Control & Integrated GNC
- Project 2 becomes the computational foundation for Projects 3–4

What Should You Build?

- Two Major Components
  1. Vehicle Simulation: Develop a physics-based numerical model including
     - Equations of motion / Coordinate systems
     - Gravity / Atmospheric environment
     - Aerodynamics / Propulsion
     - Mass / inertia variation
     - Vehicle configuration changes
  2. Navigation System: Generate realistic sensor measurements and estimate
     - Position / Velocity
     - Attitude / Angular velocity
     - Other states required for GNC

Do not simply reproduce the trajectory.
Build the physics that generates the trajectory.

Simulation & Validation

- Demonstrate That Your Simulator Works
  - Select representative flight phases
- Examples
  - Powered ascent, Engine cutoff / stage separation, Coast, Boostback, Atmospheric r eentry, Terminal descent, Landing
- **Compare Key Results** (Plot, when applicable):
  - Altitude vs. time / Downrange distance vs. time / Velocity vs. time
  - Flight-path angle / Attitude and angular rates / Dynamic pressure
  - Angle of attack / Mass and propellant
  - Navigation errors
- Validation
  - Compare your results with publicly available flight data

Model Fidelity & Engineering Judgment

- Use Available Data and Engineering Judgment
  - Not every vehicle parameter will be publicly available
  - Clearly distinguish:
    - Known: Official or reliable published information
    - Estimated: Derived from Public flight data / Videos / Figures Technical publications
    - Assumed: Engineering assumptions required to complete the model

Deliverables, Presentation & Evaluation

- Deliverables
  - Each team will submit:
    - Report – Manual-style summary, Simulation results, Validation and error analysis, List of assumptions and data sources
    - Simulation architecture
    - Vehicle dynamics model
    - Environment / propulsion / aerodynamic models
    - Navigation architecture
    - Sensor models
    - Navigation algorithm
    - Simulation code
  - Due date: 22:00 PM, November 10 (Tuesday)

Evaluation

Category	Weight
Vehicle dynamics modeling	15%
Environment & subsystem modeling	15%
Navigation system design	25%
Simulation implementation	15%
Verification & validation	10%
Technical reasoning & assumptions	10%
Report quality	10%

- Final Question
  - If this simulator were used to design the vehicle's Guidance and Control algorithms, which modeling errors or navigation errors would be most dangerous?

Teams

Team members

Team 1	Team 2	Team 3	Team 4	Team 5
Julide	Otto	Julius	Jonas	Philipp
Taeyoon	Jiwon Bae	Wonho	Seongjae	Kwanwoo
Jaehyun	Vasco	Gyeongsup	Anand	Emillien
Junghong	Jiwon Park			